

23. (2.75 points)

A multiline insurance company is considering different reinsurance options. The following table contains the cumulative probability distributions for net premium minus net losses under three scenarios: BARE (no reinsurance), the current reinsurance structure, and the alternative reinsurance structure.

Probability	BARE	Current	Alternative
0.00%	-217,911,928	-196,110,272	-154,691,590
0.25%	-71,606,742	-53,319,764	-52,785,091
0.50%	-50,326,877	-31,517,432	-26,643,103
1.00%	-31,241,388	-14,127,673	-11,094,195
2.00%	-17,530,205	-5,375,766	-6,715,733
4.00%	-8,476,413	-2,672,775	-4,229,923
8.00%	-2,252,408	-1,431,366	-2,746,629
10.00%	-559,293	-1,084,345	-2,028,415
20.00%	3,431,549	1,100,974	1,980,256
25.00%	4,197,741	1,929,228	2,993,689
50.00%	5,519,858	3,043,212	4,453,299

The following table shows the ceded premium and expected recoveries under each reinsurance structure after reinstatement:

	Ceded Premium	Expected Recoveries
Current	4,307,000	2,632,000
Alternative	2,700,000	1,676,000

a. (1.75 points)

Construct a cost/benefit diagram at each of the following return periods:

- i. 1-in-10 years
- ii. 1-in-25 years
- iii. 1-in-100 years
- iv. 1-in-200 years

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(23 continued)

b. (0.75 point)

Recommend a reinsurance structure for the company to implement utilizing the results in part a. above.

c. (0.25 point)

The model that produced the loss distributions above used a Frank copula for correlating lines of business for clash and accumulation events. Briefly describe a possible improvement to the model.

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QUESTION 23

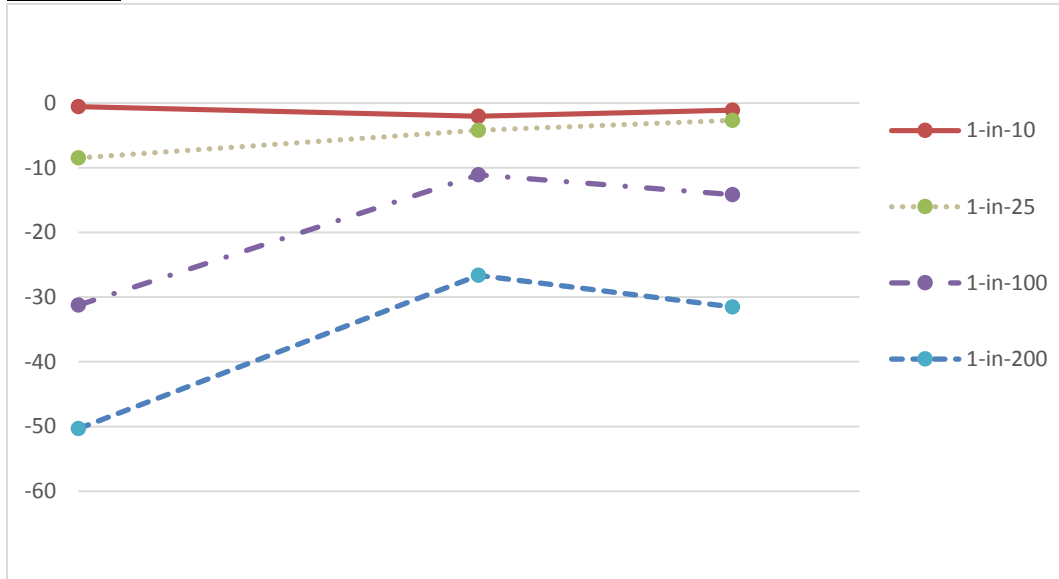
TOTAL POINT VALUE: 2.75

LEARNING OBJECTIVE(S): C2 and C6

SAMPLE ANSWERS

Part a: 1.75 points

Sample 1



Vertical axis was labeled "Net Prem – Net Loss (M)"

Horizontal axis was labeled "(000) Prem – Recov"

At the bottom of the graph, the three positions below the sets of dots were labeled:

- 0 (bare)
- 1024 (alt)
- 1675 (curr)

Sample 2

Net cost

Bare = 0

Current = 4307-2632 = 1675

Alt = 2700 – 1676 = 1024

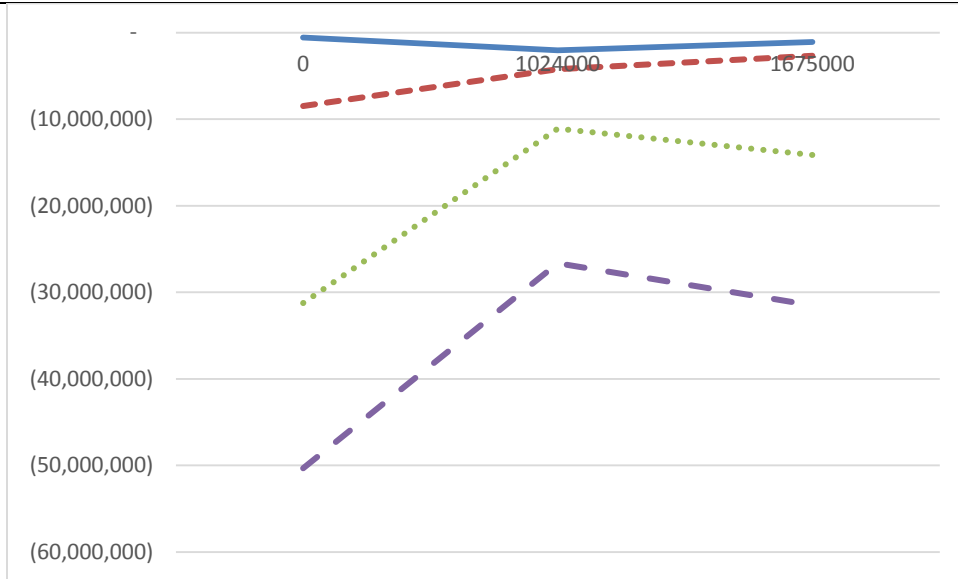
1 in 10 -> 10% prob level

1 in 25 -> 4% prob level

1 in 100 -> 1% prob level

1 in 200 -> .5% prob level

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X axis was labeled "cost"

Y axis was labeled "Prem-Loss (benefit)"

Sample 3

Benefit is decrease in losses compared to no reinsurance, cost is premium – recoveries

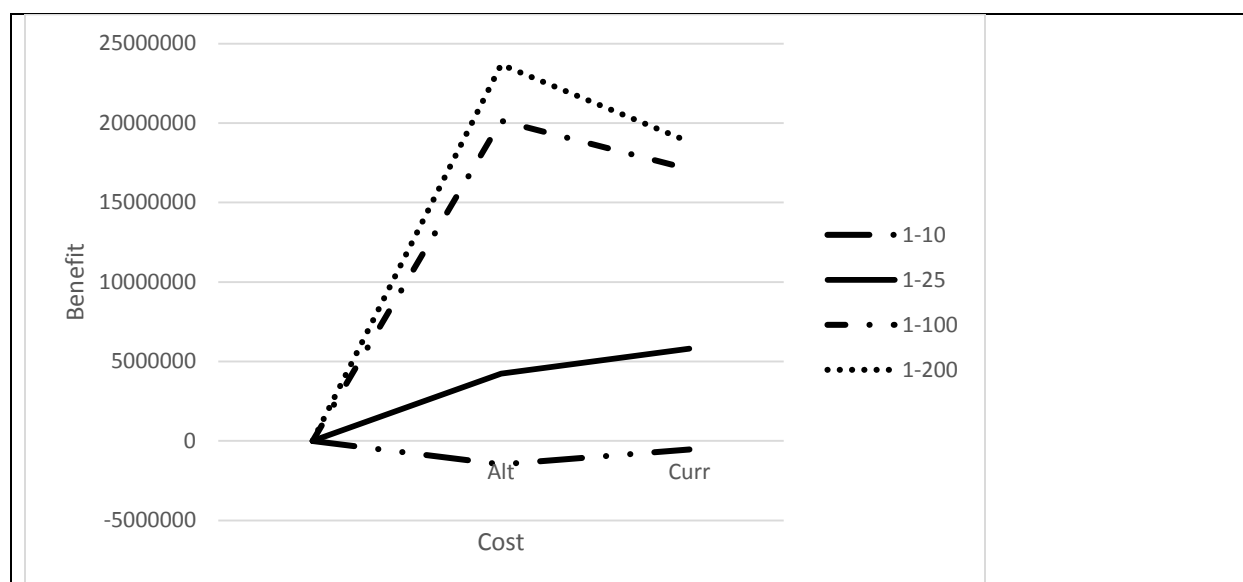
Prob	Benefit-Curr	Benefit-Alt
1%	-14,127,673- -31,211,388 = 17,113,715	20,147,193
4%	5,803,641	4,246,190
10%	-525,052	-1,469,122
.5%	18,809,445	23,683,774

Cost for Bare = 0

Cost Curr = 4,307,000 – 2,632,000 = 1,675,000

Alt = 1,024,000

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Part b: 0.75 point

Sample 1

Recommend the alternative strategy since we have lower expected risk at a lower cost, especially evident at the 1-in-100 & 1-in-200 returns.

Sample 2

Alternative has worse results at 1 in 10 and 1 in 25 levels, but better results at 1 in 100 and 1 in 200 levels.

I recommend the alternative program as it provides more protection for extreme events and has a lower net cost. Would not choose bare because it provides no protection from extreme events.

Part c: 0.25 point

Sample 1

We could use the Gumbel or Heavy Right tail copula to better fit the right tail correlation seen in this model. These 2 copulas have more heavily correlated tails.

Sample 2

The Frank copula has little correlation in the right tail (0 when $Z=1$). For accumulation and clash events, we want correlation in right tail; use Gumbel copula instead to correlate.

Sample 3

Could use a copula that has greater correlation in the right tail, as these 2 LOB are likely highly correlated in the tail. Use the heavy right tail copula instead since it has greater correlation in the right tail.

EXAMINER'S REPORT

Candidates were expected to demonstrate an understanding of cost-benefit analysis as it applies to evaluating competing reinsurance programs, and were expected to know properties of various copulas and how appropriate those properties are for modeling reinsurance losses.

Part a

Candidates were expected to:

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- Derive the cost of the Alternate and Current reinsurance programs.
- For each of the four percentiles, chart expected benefit against cost for all three reinsurance programs.

Common errors include:

- Displaying the four cost-benefit charts on separate graphs, without maintaining a consistent scale that would allow comparisons of the relative benefits of the three programs.
- Charting expected benefit against distribution percentile, instead of against cost.
- Not including the Bare program in their diagram. Some candidates performed the relevant subtractions to find the cost and benefit of the Alternate and Current programs relative to Bare, but to earn full credit, the candidate had to then identify the origin on their diagram as the Bare program.
- Not understanding that the net-premium-minus-net-losses distribution represented the (relative) benefit, while ceded premium minus expected recoveries was the cost.

Part b

Candidates were expected to identify the Alternate reinsurance program as the best of the three options, and mention both the lower cost and the higher protection in the tail of the distribution as supporting reasons.

Common errors:

- Selecting the Current program because the far tail was too unlikely or too difficult to model.
- Misunderstanding that the reinsurance structure cannot be selected after a given scenario has come to pass.

Part c

Candidates were expected to describe at least one way their proposed model improvement addressed a weakness in the model presented. This generally entailed providing support for the choice of a copula that provided more tail correlation than the Frank copula.